Sign Charts, quickly

Winston Cheong

September 2023

1 The method

Given a rational function, put it into factored form. Use the factored form to split up a number line. The sign of each interval can be quickly determined from this factored form.

Starting at the right, all factors should be positive, so the sign of the right-most interval is determined by the outer sign.

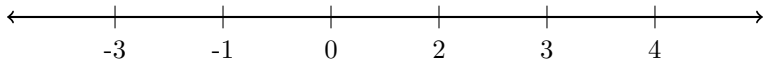
Now, as you move to the left, you cross a tick mark. The corresponding factor will flip in sign. The sign of the interval only depends on the power of that factor (even power means the sign doesn't flip on the number line; odd power means the sign flips on the number line). Continue until you are done.

2 Worked example

Consider

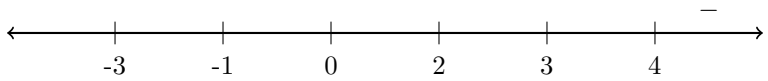
$$\frac{-4(x-3)^7(x-2)^8(x+3)^4}{x^6(x-4)^5(x+1)^3}$$

The tick marks are $x = 3, 2, -3, 0, 4, -1$:



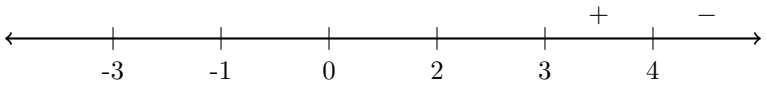
Starting at the right, each factor is positive. The "-4" in the front makes the sign '-':

$$\frac{-4(+)^7(+)^8(+)^4}{(+)^6(+)^5(+)^3}$$



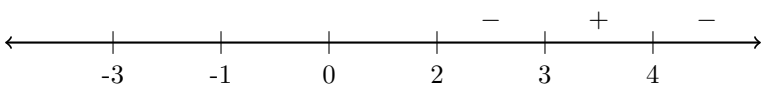
Moving to the left, we first cross $x = 4$, which corresponds to the factor $(x-4)$. This flips the factor. It's power is 5, so the sign *does* flip on the number line.

$$\frac{-4(x-3)^7(x-2)^8(x+3)^4}{x^6(x-4)^5(x+1)^3}$$



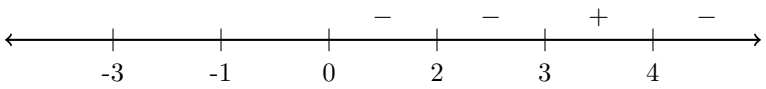
Next, we cross $x = 3$. It is similar:

$$\frac{-4(x-3)^7(x-2)^8(x+3)^4}{x^6(x-4)^5(x+1)^3}$$



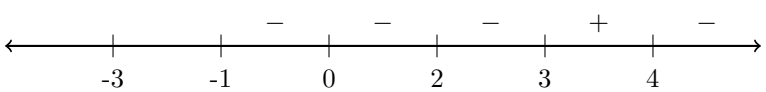
Next we cross $x = 2$. This time, the corresponding factor $(x-2)$ is raised to an even power. Although $x-2$ will flip in sign as we cross that tick mark, the factor $(x-2)^8$ will not, so the sign doesn't flip on the number line:

$$\frac{-4(x-3)^7(x-2)^8(x+3)^4}{x^6(x-4)^5(x+1)^3}$$

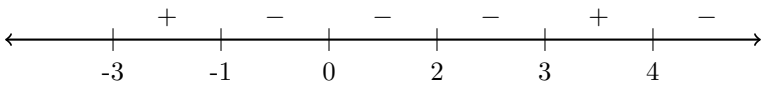


Doing the rest in quick succession:

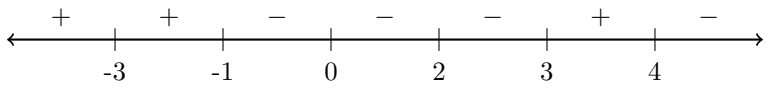
$$\frac{-4(x-3)^7(x-2)^8(x+3)^4}{x^6(x-4)^5(x+1)^3}$$



$$\frac{-4(x-3)^7(x-2)^8(x+3)^4}{x^6(x-4)^5(x+1)^3}$$



$$\frac{-4(x-3)^7(x-2)^8(x+3)^4}{x^6(x-4)^5(x+1)^3}$$



3 Some practice

Draw the sign chart for the following functions:

1. $3x(x-1)(x-2)^2(x-3)^3$
2. $\dfrac{x^2-5x-6}{(x+1)^4}$
3. $\dfrac{1-x^2}{(1+x^2)^2}$
4. $\dfrac{2x(x^2-3)}{(x^2+1)^3}$